



SERANGOON JUNIOR COLLEGE  
JC2 PRELIMINARY EXAMINATION 2011

BIOLOGY  
Higher 1



Thursday  
25 August 2011

1 hour

Additional materials:  
Optical Mark Sheet

### INSTRUCTIONS TO CANDIDATES

Write your name, CG and index number in the spaces at the top of this page.

Enter your name, subject title, test name, class. For your index number, enter your full NRIC number. Shade the corresponding lozenges on the OMS according to the instructions given by the invigilators.

There are forty questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C, D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the OMS.

**AT THE END OF THE EXAMINATION, HAND IN BOTH OMS AND QUESTION PAPERS.**

### INFORMATION FOR CANDIDATES

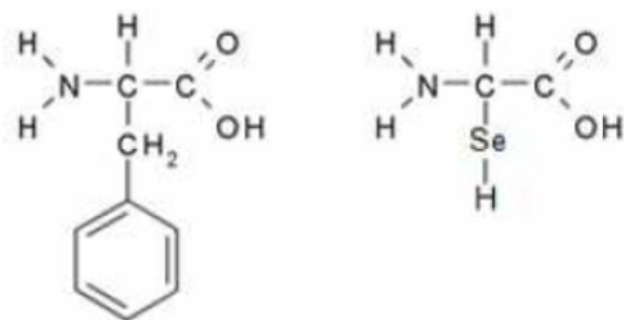
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done on the question paper.

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**This question paper consists of 17 printed pages.**

1. The diagram shows two molecules of a biological substance.



Which type of bond is formed when these molecules react to form a larger molecule?

- A Phosphodiester
- B Glycosidic
- C Peptide
- D Hydrogen

2. Which one of the following statements is **INCORRECT**?

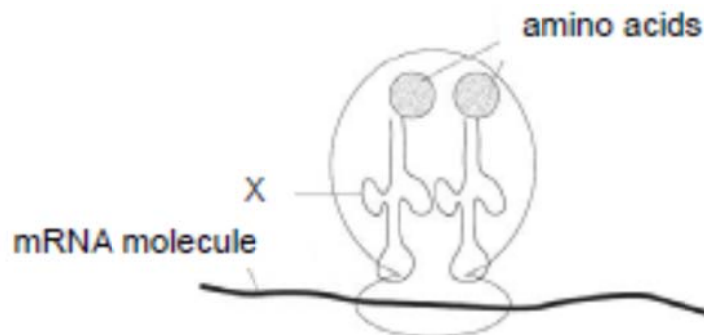
- A There are 3 polypeptide chains in the collagen helix, and only one in the  $\alpha$ -helix.
- B Proline is a major component of collagen but not of the  $\alpha$ -helix.
- C Every third residue in the  $\alpha$ -helix must be glycine, but there is no sequence regularity in collagen
- D Some hydrogen bonding exists between different chains in collagen, but hydrogen bonding occurs within the same chain in the  $\alpha$ -helix.

3. Which of the following is/ are true of triglycerides and phospholipids?

- I. Both contain fatty acid tails
- II. Only phospholipids are amphipathic
- III. Phospholipids are formed from fatty acids and phosphoric acid while triglycerides are formed from fatty acids and glycerol.
- IV. Both fatty acids and glycerol are soluble in organic solvents
- V. Phospholipids have a higher carbon content than triglycerides for storage purposes

- A I, II and III
- B I, II and IV
- C II, III and IV
- D II, III and V

4. The diagram shows part of a molecule of mRNA bound to a ribosome.



Which of the following is/are characteristic(s) of molecule X?

- I It will form base-pairs via hydrogen bonds only with mRNA.
- II It is formed by RNA polymerase in the nucleus.
- III It will be attached to amino acid methionine if it carries anticodon AUG.
- IV It is a substrate for amino acyl-tRNA synthetase.

- A I, III only
- B I, II, III only
- C II, IV only
- D I, II, IV only

5. Enzyme molecules with high optimum temperature will have

- A greater number of peptide bonds than other enzyme molecules.
- B greater number of cysteine residues in their polypeptide chains.
- C greater number of hydrophobic amino acid residues.
- D greater number of associated non-protein groups.

6. Which of the following property of enzymes is **NOT** directly related to protein structure?

- I Specificity in the reaction catalysed
- II Sensitivity to pH changes
- III Increased action over a temperature range of 0°C to 30°C
- IV Inhibition by high temperature

- A I only
- B III only
- C II and III
- D I, II and IV

7. A pear was cut into six equal segments, two of which were blanched by immersing them in boiling water for one minute. Each segment of the pear was then cut into pieces and these were treated in several ways as indicated in the table below. The time taken for each piece of pear to develop a standard degree of browning was then recorded. You may assume that all pieces had the same initial colour

| Liquid in which pear pieces were immersed | Initial treatment of pear segment | Time for appearance of browning at the temperature indicated (min) |      |       |
|-------------------------------------------|-----------------------------------|--------------------------------------------------------------------|------|-------|
|                                           |                                   | 20°C                                                               | 40°C | 100°C |
| Water                                     | Unblanched                        | 25                                                                 | 9    | X     |
|                                           | blanched                          | X                                                                  | X    | X     |
| 1% glucose solution                       | Unblanched                        | X                                                                  | X    | 5     |
|                                           | blanched                          | X                                                                  | X    | 4     |
| 0.1M nitric acid                          | Unblanched                        | X                                                                  | X    | X     |
| 2% dithionite solution                    | Unblanched                        | X                                                                  | 135  | X     |

X indicates that the piece of pear did not turn brown within 180 minutes.

The enzyme involved in the browning of the pear pieces is

I thermostable.

II destroyed by blanching.

III inhibited more by 2% dithionite solution than by 0.1M nitric acid.

IV is not the only agent responsible for the browning of the pear in the experiment.

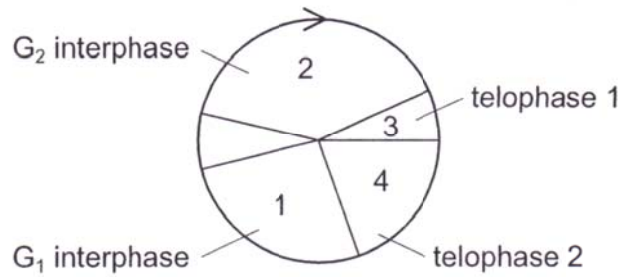
A I and III only

B II and III only

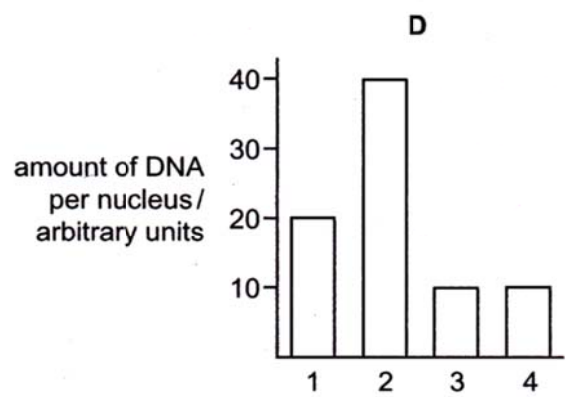
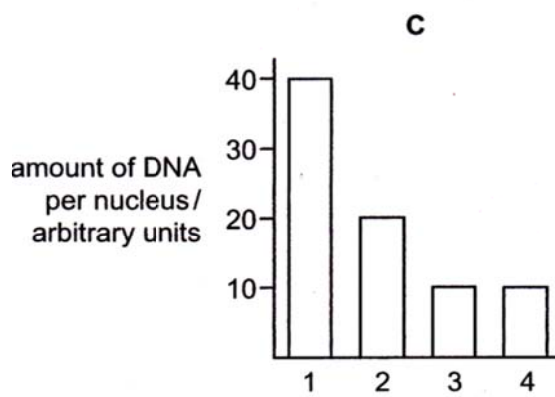
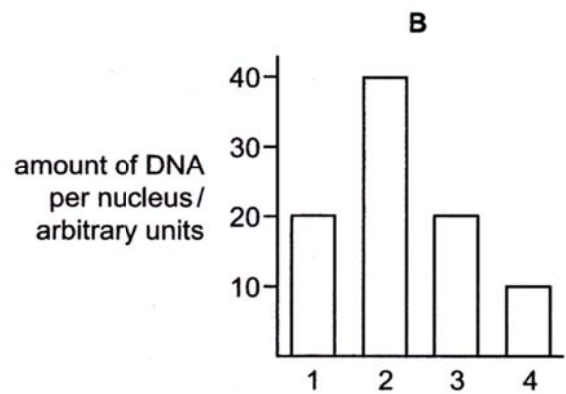
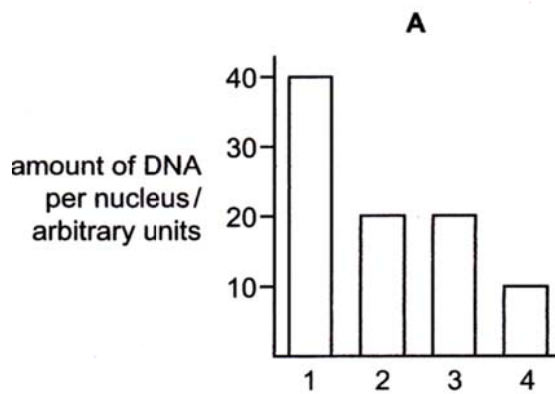
C II and IV only

D II, III and IV only

8. The amount of DNA per nucleus in a cell during a meiotic cell cycle, as shown below, was measured.

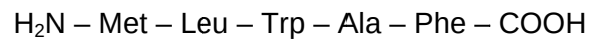


Which bar graph correctly represents the variation in the DNA content?



9. During gametogenesis, non-disjunction occurs in one daughter cell during meiosis II. What would be the result at the completion of meiosis?
- A** All gametes would be haploid.
  - B** Two gametes would be haploid and two will be diploid.
  - C** Two gametes would have  $(n+1)$  chromosomes and two would have  $(n-1)$  chromosomes.
  - D** One gamete would have  $(n+1)$  chromosomes, one would have  $(n-1)$  chromosomes and the other two will have  $n$  chromosomes.
10. In a hypothetical species of frogs, there are 6 tetrads at metaphase of meiosis I. How many chromosomes would be present in each cell at the end of meiosis II?
- A** 3
  - B** 6
  - C** 12
  - D** 24
11. Identify the correct statements regarding the replication of DNA
- 1 Only 1 strand of DNA is replicated
  - 2 It occurs in S phase of interphase in both mitosis and meiosis
  - 3 It occurs after the completion of telophase in meiosis
  - 4 DNA polymerase III is required (for bacteria)
- A** 1 and 3 only
  - B** 2 and 4 only
  - C** 2 and 3 only
  - D** 2 only
12. A synthetic peptide contains ten amino acids of four different kinds. What is the theoretical minimum number of tRNA molecules required to translate the mRNA for this synthetic peptide?
- A** 4
  - B** 10
  - C** 12
  - D** 40

13. A pentapeptide has the following amino acid sequence:



where

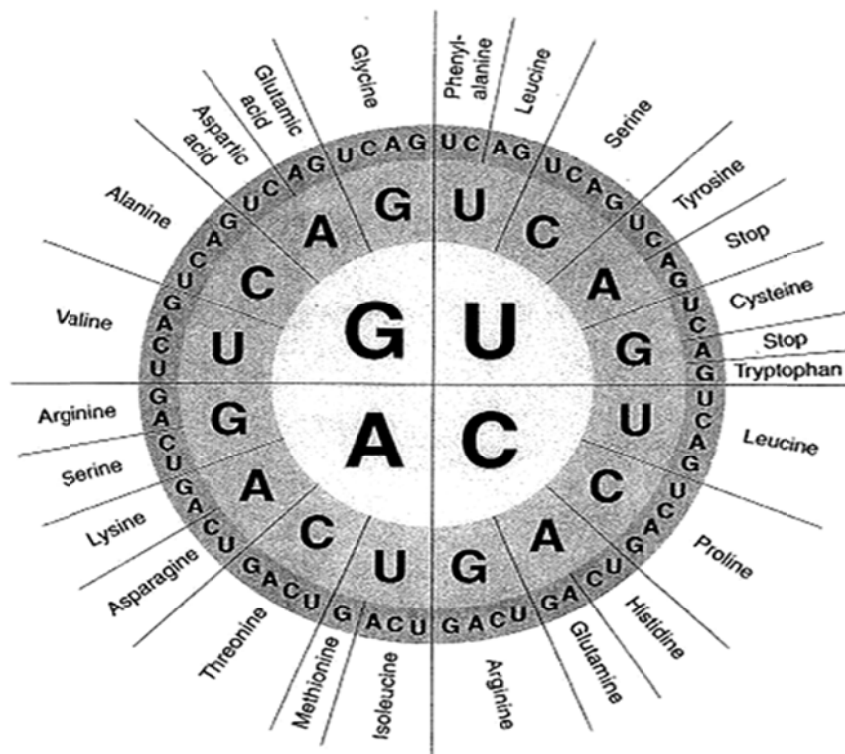
Met : methionine

Ala : alanine

Leu : leucine

Phe : phenylalanine

Trp : tryptophan



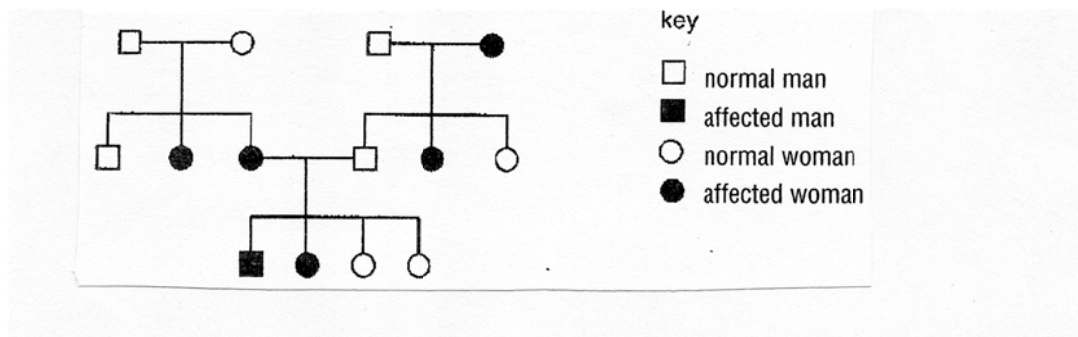
Using the genetic code shown in the diagram above, deduce the corresponding template strand sequence during transcription:

- A 5' – ATG TTG TGG GCC TTT TAG – 3'
- B 5' – AUG CUG UGG GCA UUC – 3'
- C 5' – TTA GAA GGC CCA CAA CAT – 3'
- D 5' – GAA GGC CCA CAA CAT – 3'

14. Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein product of that gene?

|          | Promoter                              | Terminator                             | Start codon                            | Stop codon                             | Middle of an intron                   |
|----------|---------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|---------------------------------------|
| <b>A</b> | No protein product is produced        | Protein product is shorter than normal | Protein product is longer than normal  | Protein product is normal              | Too much protein product is produced  |
| <b>B</b> | Too much protein product is produced  | Protein product is normal              | No protein product is produced         | Protein product is longer than normal  | Protein product is normal             |
| <b>C</b> | Protein product is normal             | Protein product is longer than normal  | Protein product is shorter than normal | Too much protein product is produced   | Protein product is longer than normal |
| <b>D</b> | Protein product is longer than normal | Too much protein product is produced   | Protein product is normal              | Protein product is shorter than normal | No protein product is produced        |

15. The family tree shows the inheritance of a skin condition.



What is the most likely genetic basis for the occurrence of this defect?

- A** An autosomal recessive
- B** An autosomal dominant
- C** Sex-linked recessive
- D** Sex-linked dominant



16. The following two examples illustrate the inheritance of a sex-linked gene on the X-chromosome. In the parental generation (*P*), the homogametic sex is homozygous for colour in both examples.

| Cat                                      | Magpie Moth                               |
|------------------------------------------|-------------------------------------------|
| (black dominant to yellow)               | (normal colour dominant to pale colour)   |
| <i>P</i> Black male x yellow female      | <i>P</i> Pale colour male x normal female |
| <i>F</i> 1 yellow male to 1 black female | <i>F</i> 1 normal male to 1 pale female   |

In these crosses, the heterogametic sex is

- A** male in the cat, female in the moth
- B** female in the cat, male in the moth
- C** male in both the cat and moth
- D** female in both the cat and moth

17. Which combination is correct?

|          | ATP used     | Dehydrogenation | Reduced NADP used |
|----------|--------------|-----------------|-------------------|
| <b>A</b> | Calvin cycle | glycolysis      | Krebs cycle       |
| <b>B</b> | glycolysis   | Calvin cycle    | Krebs cycle       |
| <b>C</b> | glycolysis   | Krebs cycle     | Calvin cycle      |
| <b>D</b> | Krebs cycle  | glycolysis      | Calvin cycle      |

18. The blue dye DCPIP can be converted to reduced DCPIP by gaining electrons. This is summarised below.



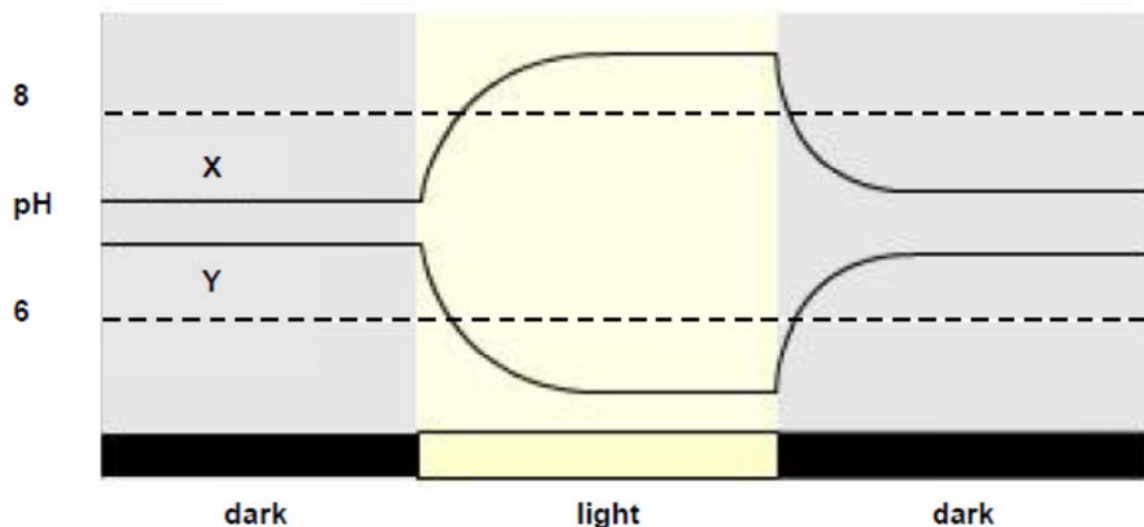
| Tube | Contents                                                             | Treatment            | Colour of solution |              |
|------|----------------------------------------------------------------------|----------------------|--------------------|--------------|
|      |                                                                      |                      | At start           | After 20 min |
| 1    | 1 cm <sup>3</sup> chloroplast suspension and 5 cm <sup>3</sup> DCPIP | Illuminated strongly | Blue green         | Green        |
| 2    | 1 cm <sup>3</sup> buffer solution and 5 cm <sup>3</sup> DCPIP        | Illuminated strongly | Blue               | Blue         |
| 3    | 1 cm <sup>3</sup> chloroplast suspension and 5 cm <sup>3</sup> DCPIP | Left in the dark     | Blue green         | Blue green   |

A suspension of chloroplasts was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were then prepared and treated in the following way.

Which one of the following statements is a possible conclusion for the observations above?

- A** Light reaction which occurs in the chloroplasts yield free electrons which reduced DCPIP.
- B** Either strong illumination or the buffer used in extraction of chloroplasts could oxidise DCPIP
- C** Photolysis of water produces oxygen which oxidises DCPIP.
- D** Respiration consumes oxygen from the suspension, hence DCPIP is partially reduced.

19. The diagram below consists of two graphs showing the changes in pH of two structures X and Y, of a chloroplast in the absence and presence of light. Which of the following correctly identifies structures X and Y?



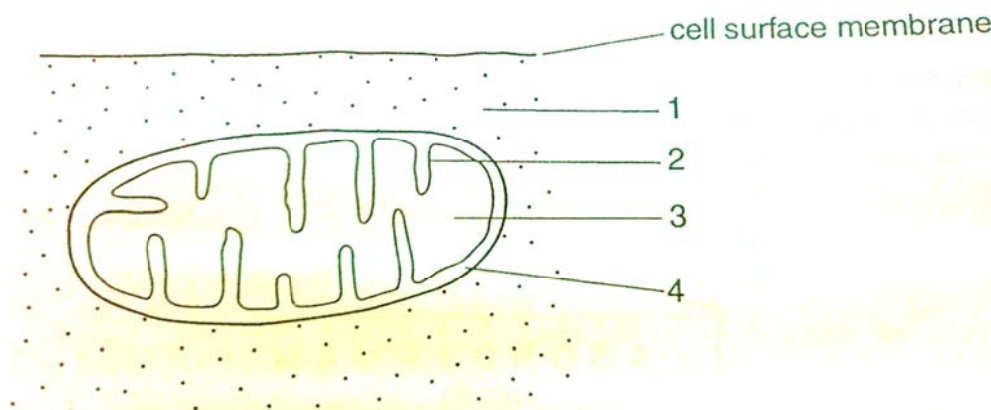
|   | X               | Y               | Reason                                                                                           |
|---|-----------------|-----------------|--------------------------------------------------------------------------------------------------|
| A | Thylakoid space | Stroma          | Protons are pumped from the thylakoid space into the stroma, hence decreasing the latter's pH    |
| B | Stroma          | Thylakoid space | Protons are pumped from the stroma into the thylakoid space, hence decreasing the latter's pH    |
| C | Thylakoid space | Stroma          | Protons are pumped from the thylakoid membrane into the stroma, hence increasing the latter's pH |
| D | Stroma          | Thylakoid space | Protons are pumped from the stroma into the thylakoid membrane, hence increasing the latter's pH |

20. Read the following statements.

I Two turns of the Krebs cycle is required to oxidize 1 molecule of glucose.  
 II Two molecules of carbon dioxide are generated for every molecule of acetyl CoA introduced into the citric acid cycle.  
 III Aerobic respiration can produce about 19 times the amount of ATP produced in anaerobic respiration.

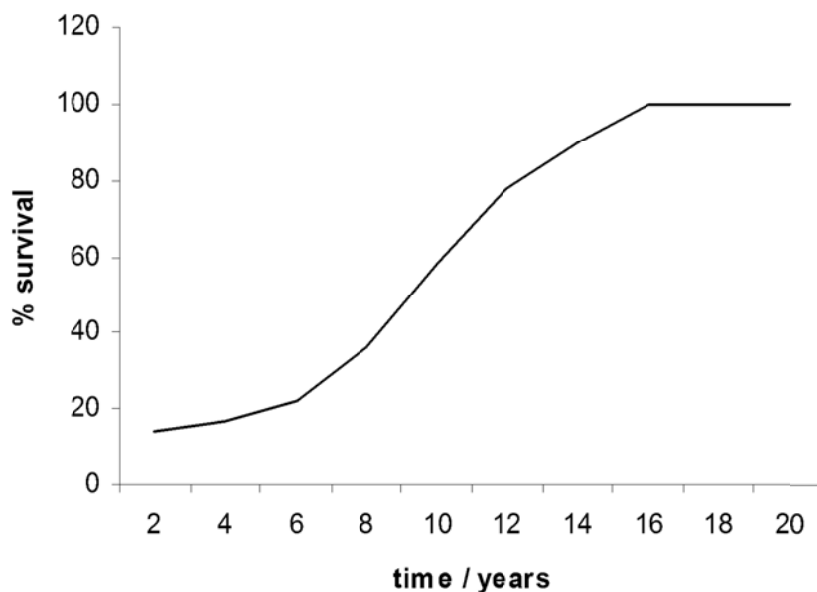
- A I only  
 B I, II and III  
 C II and III only  
 D I, and II only

21. The diagram shows a mitochondrion in a cell.



Where is ATP produced from ADP during respiration?

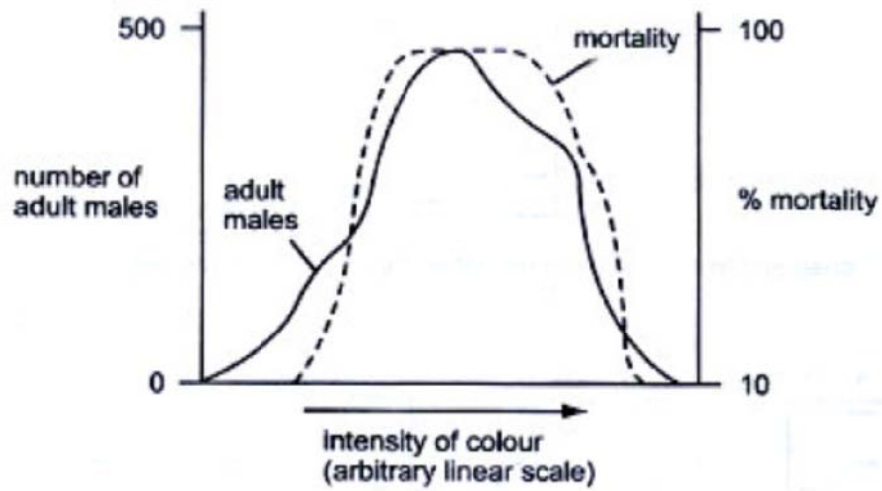
- A 1 and 2 only
  - B 1, 2 and 3
  - C 2 and 3 only
  - D 2, 3 and 4
22. The graph shows the effect of pesticide treatment on beetles in Tanzania over a number of years. A standard amount of pesticide was used each year in summer.



How is the effect of pesticide best explained?

- A A few resistant beetles reproduced more successfully and the resistance allele increased in frequency.
- B At every generation, an increasing proportion of beetles mutated to become resistant.
- C Repeated exposure to the pesticide caused the beetles to become more resistant.
- D The allele for resistance mutated from the recessive form to the dominant form

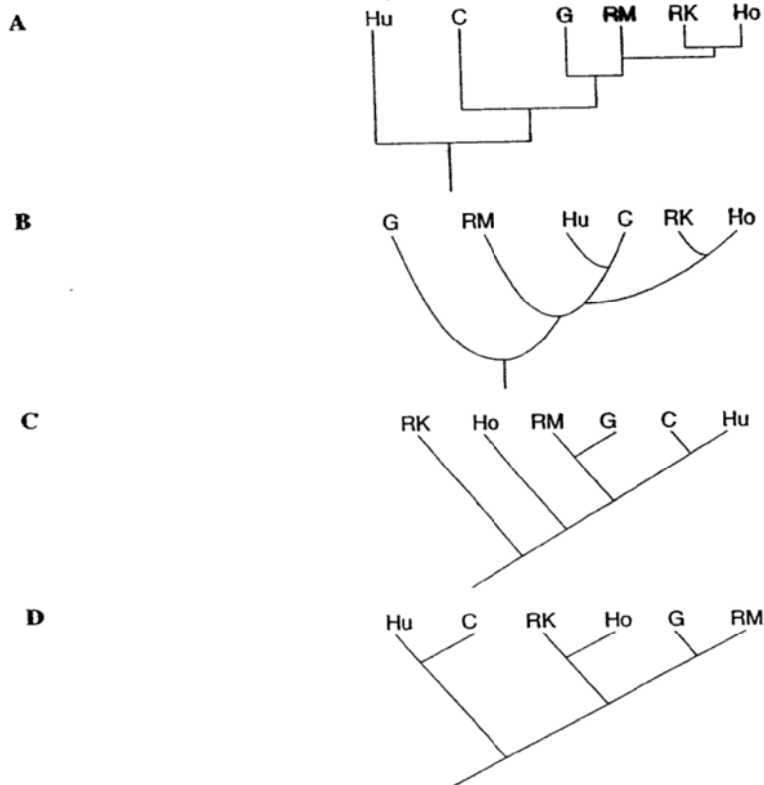
23. The graph below shows data on a population of a species of moth which shows considerable variation in colour intensity. Which conclusion can be made from this graph?



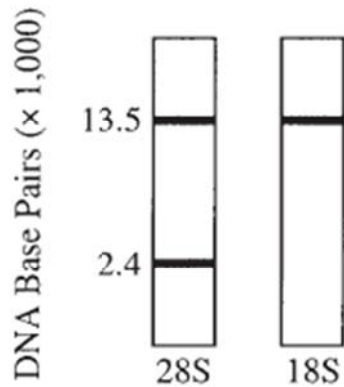
- A Colour variation is environmentally induced.
- B Colour variation is genetically determined.
- C Extreme forms are favoured by natural selection.
- D The species shows discontinuous variation with respect to colour.

24. The table below shows the amino acid sequence of part of the beta-haemoglobin molecule found in 6 different vertebrates. The beta-globin chain has a total of 146 amino acids. Only the amino acids found in positions 101 to 116 are represented.

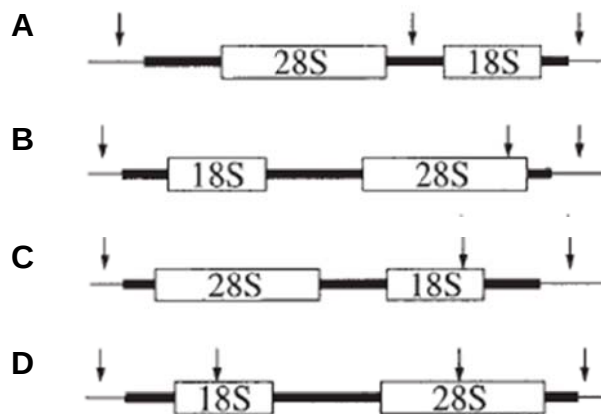
| amino acid position | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 | 111 | 112 | 113 | 114 | 115 | 116 |
|---------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| human (Hu)          | glu | asn | phe | arg | leu | leu | gly | asn | val | leu | val | cys | val | leu | ala | his |
| red kangaroo (RK)   | glu | asn | phe | lys | leu | leu | gly | asn | ile | ile | val | ile | cys | leu | ala | glu |
| rhesus monkey (RM)  | glu | asn | phe | lys | leu | leu | gly | asn | val | leu | val | cys | val | leu | ala | his |
| gorilla (G)         | glu | asn | phe | lys | leu | leu | gly | asn | val | leu | val | cys | val | leu | ala | his |
| chimpanzee (C)      | glu | asn | phe | arg | leu | leu | gly | asn | val | leu | val | cys | val | leu | ala | his |
| horse (Ho)          | glu | asn | phe | arg | leu | leu | gly | asn | val | leu | ala | leu | val | val | ala | arg |



25. The autoradiograms obtained below (after electrophoresis and Southern Blotting) show human DNA digested with a specific restriction enzyme and probed with labeled rRNA. In the autoradiogram on the left, the probe was 28S rRNA; at the right, the probe was 18S rRNA.



If the arrow in the following map shows the location of the restriction sites of this restriction enzyme, which map best explains the result shown above?



26. Genetically engineering techniques may be used to mass produce human proinsulin using E.coli as the expression system.

Some events that take place during the process are listed below.

- I      Insert the plasmid vector into a plasmid-free E.coli.
- II     Use a restriction endonuclease to cut the plasmid vector at specific points.
- III    Purify proinsulin from the E.coli.
- IV    Grow the E.coli in an industrial fermenter.
- V      Insert the proinsulin gene into the plasmid vector.

Which sequence of events is correct?

first → last

- A**    II, I, V, III, IV
  - B**    II, I, V, IV, III
  - C**    II, V, I, III, IV
  - D**    II, V, I, IV, III
27. Which of the following statements are true about all stem cells?
- 1. Stem cells can be induced to differentiate by signal molecules.
  - 2. Stem cells are easily isolated and propagated.
  - 3. Stem cells are able to make more stem cells.
  - 4. Adult stem cells can be derived from umbilical cord.
- A**    1 and 2
  - B**    1 and 3
  - C**    1,2 and 3
  - D**    2,3 and 4
28. What is **not** a possible issue of concern over the creation of genetically modified farmed animals?
- 1. Genetic engineering may result in the creation of new proteins that are harmful to the organisms that produce or consume them.
  - 2. Cross species gene transfer may compromise the genome integrity of the species involved.
  - 3. Over production of certain gene products may cause undue stress to the genetically modified farmed animals.
  - 4. Some genetically modified food products may not be acceptable to certain groups of people.
- A**    1 and 4
  - B**    2 and 3
  - C**    1,3 and 4
  - D**    none of the above



29. Which statements about the Human Genome Project (HGP) are true?
- 1: The entire human genome has already been sequenced.
  - 2: Mapping of the genome is not required once the entire sequence of the human genome has been obtained.
  - 3: The goal of the HGP is also to address ethical issues such as genetic privacy.
  - 4: Since the human genome has already been sequenced, the money invested in the HGP has been fully justified.
- A** 1 and 3  
**B** 2 and 3  
**C** 1,2 and 4  
**D** all of the above
30. Which feature of stem cells obtained from blood in the umbilical cord enables their use in the treatment of a variety of blood cancers?
- A** They can differentiate into bone marrow cells.  
**B** They can differentiate into any cellular component of blood.  
**C** They can replace blood stem cells affected by cancer.  
**D** They are totipotent

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